# AN ALGEBRAIC SPACE WHOSE REDUCTION IS A SCHEME IS A SCHEME (THICKENING FORM)

THE STACKS PROJECT AUTHORS

*Editorial note.* Statement span is lines 1729-1733 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 1735-1742, the complete author proof (6 body lines).

*Editorial note.* Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

*Editorial note.* Text is retained verbatim, including the author's own typos and placeholders.

*Editorial note.* Only the author's FIRST proof (lines 1735-1742) is retained; the chapter's separate direct proof for finite order thickenings lies outside the collected spans.

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. The Limits of Spaces lemma equating being-a-scheme with being-a-scheme after reduction (spaces-limits-lemma-reduction-scheme) and the invariance of the reduction under thickenings; cited inputs with the argument written in the retained span.

*Source reference.* Original lemma 07VU: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 1742–1742. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

**Lemma 0.1.** *Let $S$ be a scheme. Let $X \subset X'$ be a thickening of algebraic spaces over $S$. If $X$ is (representable by) a scheme, then so is $X'$.*

*Proof.* Note that $X'_{red} = X_{red}$. Hence if $X$ is a scheme, then $X'_{red}$ is a scheme. Thus the result follows from Limits of Spaces, Lemma 07VU. Below we give a direct proof for finite order thickenings which is the case most often used in practice. □

*Date*: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.